

User Guide - LLM / VLM Inference Platform

This guide covers how to call the platform once you have an API key. All requests are plain HTTP (OpenAI-compatible), non-streaming.

1. Endpoint

```
http://10.208.211.62:18000/v1/chat/completions
```

Always bypass the corporate proxy for this internal address:

```
curl --noproxy '*' ...
```

WINDOWS USERS: the examples below use bash-style single quotes. If running from cmd.exe (not Git Bash / WSL), single quotes are NOT string delimiters there - your JSON body will silently arrive empty, giving back {"detail":[{"type":"json_invalid", ..., "input":{}}]}. On cmd.exe, use double quotes around the whole -d payload and escape inner double quotes with backslash:

```
curl --noproxy "" -s -X POST -H "Authorization: Bearer <YOUR_API_KEY>" -H "Content-Type: application/json" -d '{"model":"ongc-llm","messages":[{"role":"user","content":"hello"}],"max_tokens":256}' http://10.208.211.62:18000/v1/chat/completions
```

PowerShell has its own quoting quirks too - if in doubt, use Git Bash/WSL and the bash examples as-is, or save the JSON body to a file and pass -d @payload.json instead of inlining it.

2. Authentication

Pass your API key as a Bearer token in the Authorization header:

```
-H "Authorization: Bearer <YOUR_API_KEY>"
```

Replace <YOUR_API_KEY> with the key you were given (starts with sk-ongc-).

3. Text request (LLM)

```
curl --noproxy '*' -s -X POST \
-H "Authorization: Bearer <YOUR_API_KEY>" \
-H "Content-Type: application/json" \
-d '{
  "model": "ongc-llm",
  "messages": [
    {"role": "user", "content": "Summarize the key risks in a slope stability assessment."}
  ],
  "max_tokens": 512,
  "temperature": 0.7
}' \
http://10.208.211.62:18000/v1/chat/completions
```

Response (trimmed):

```
{
  "model": "ongc-llm",
  "choices": [
    {"message": {"role": "assistant", "content": "..."}}
  ],
  "usage": {"prompt_tokens": 14, "completion_tokens": 238, "total_tokens": 252}
}
```

4. Image request (VLM)

To ask a question about an image, include an image_url content part in the message alongside your text. The platform automatically detects this and routes it to the vision model - no separate endpoint or flag needed.

```
curl --noproxy '*' -s -X POST \
-H "Authorization: Bearer <YOUR_API_KEY>" \
-H "Content-Type: application/json" \
-d '{
  "model": "ongc-llm",
  "messages": [
    {
      "role": "user",
      "content": [
        {"type": "text", "text": "What rock formation is visible in this image?"},
        {"type": "image_url", "image_url": {"url":
"data:image/jpeg;base64,<BASE64_IMAGE_DATA>"}]}
      ]
    }
  ],
  "max_tokens": 512
}' \
http://10.208.211.62:18000/v1/chat/completions
```

image_url.url accepts either:

- A data:image/...;base64,<...> inline data URI (most common), or
- A reachable http(s):// URL the gateway can fetch directly.

To base64-encode a local image file:

```
base64 -w0 photo.jpg
```

5. Request parameters

Field	Type	Default	Notes
model	string	ongc-llm	Fixed alias - routing to the right backend (text/vision) is automatic and transparent. Always send this value.
messages	array	required	List of {role, content}. role is user, assistant, or system. content is a string for plain text, or a list of {type: text image_url, ...} parts for image requests.
max_tokens	int	256	Maximum tokens to generate in the response. Raise this for longer answers.
temperature	float	0.7	Higher = more random/creative; lower (e.g. 0.1-0.3) = more deterministic/focused. Use low values for factual/technical Q&A.
top_p	float	engine default	Nucleus sampling cutoff. Lower values (e.g. 0.8) narrow the model to more likely tokens.
top_k	int	engine default	Restricts sampling to the top-K most likely tokens. Common value: 20.
presence_penalty	float	engine default	Penalizes tokens that already appeared, reducing repetition. Typical range 0-2.
enable_thinking	bool	false	Convenience flag for extended step-by-step reasoning before the final answer. See Thinking mode section before turning this on.
chat_template_kw args	object	none	Advanced/raw passthrough to the model's chat template, e.g. {"enable_thinking": false}. Overrides the enable_thinking flag above.

Multi-turn conversation: include prior turns in messages in order (user, assistant, user, ...) - the platform does not retain history between requests, so the full conversation must be resent each call.

6. Thinking mode (enable_thinking)

Some requests benefit from the model reasoning step-by-step before answering (e.g. multi-step calculations or logical analysis). This is off by default. To turn it on:

```
curl --noproxy '*' -s -X POST \
-H "Authorization: Bearer <YOUR_API_KEY>" \
-H "Content-Type: application/json" \
-d '{
```

```
    "model": "ongc-llm",
    "messages": [{"role": "user", "content": "A borehole log shows 3 fault zones at depths of
120m, 340m, and 510m. What is the average spacing between them?"}],
    "max_tokens": 1024,
    "enable_thinking": true
}' \
http://10.208.211.62:18000/v1/chat/completions
```

Equivalent raw form (matches the OpenAI SDK `extra_body` style, if calling via the `openai` Python client instead of `curl`):

```
chat_response = client.chat.completions.create(
    model="ongc-llm",
    messages=messages,
    max_tokens=1024,
    temperature=0.7,
    top_p=0.8,
    presence_penalty=1.5,
    extra_body={
        "top_k": 20,
        "chat_template_kwargs": {"enable_thinking": False},
    },
)
```

Important: when thinking mode is on, set `max_tokens` generously (1024+). The reasoning itself consumes tokens from your `max_tokens` budget - if it's too low, the model can use up the entire budget reasoning and return an empty final answer.

7. Common errors

HTTP status	Meaning	What to do
401	Missing/invalid API key	Check the Authorization: Bearer ... header.
429	Rate limit exceeded	You're sending requests faster than your per-minute quota. Wait and retry (a Retry-After header is included).
503	Server busy (queue full)	
500	Internal error	Transient - retry. If it persists, report it.

8. Quick checklist

- Use `--no-proxy '*'` on every `curl` call.
- Send Authorization: Bearer <key> on every request.
- Always set "model": "ongc-llm".
- For images, base64-encode and embed as a data: URI inside content.
- Resend full conversation history each call (no server-side memory).
- Only set `enable_thinking: true` for genuinely hard, multi-step problems, and raise `max_tokens` (1024+) when you do.